IMAT 모의고사 1회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> For decades, hospital wards relied on fixed visiting hours, partly from a belief that visitors brought infections and disturbed rest. Recent studies of intensive care units that adopted flexible visiting have found no increase in infection rates and, in several cases, a reduction in patient delirium. Critics note that most of these studies were small and that staff workload was rarely measured. Nonetheless, several national health services now recommend that restrictions be justified case by case rather than imposed by default.

1. Which of the following can be most reliably inferred from the passage?

A. The original justification for fixed visiting hours was not supported by the recent studies.
B. Flexible visiting reduces infection rates in intensive care units.
C. National health services have banned fixed visiting hours.
D. Delirium is caused mainly by the absence of visitors.
E. Staff workload increases when visiting hours become flexible.

2. Which statement, if true, would most weaken the critics' position described in the passage?

A. A large multi-centre trial measuring staff workload found no significant change after flexible visiting was introduced.
B. Infection rates in hospitals have fallen over the last decade for unrelated reasons.
C. Delirium is more common in elderly patients.
D. Visiting rules differ between countries.
E. Some patients prefer fewer visitors.

3. Which scientist is credited with first describing the circulation of blood pumped by the heart around the body in the 17th century?

A. Andreas Vesalius
B. Louis Pasteur
C. Edward Jenner
D. Robert Koch
E. William Harvey

4. The Treaty of Rome (1957) is best known for establishing which organisation?

A. The United Nations
B. The World Health Organization
C. NATO
D. The European Economic Community
E. The Council of Europe

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. All surgeons in the hospital are doctors. Some doctors in the hospital speak Italian. No one who speaks Italian works night shifts. Which of the following must be true?

A. Some doctors do not work night shifts.
B. No surgeon works night shifts.
C. All doctors who work night shifts are surgeons.
D. Some surgeons work night shifts.
E. Some surgeons speak Italian.

6. A pharmacy sells a drug at €40 per box. In a promotion, the price is reduced by 25%; after the promotion, the reduced price is raised by 25%. What is the final price?

A. €38.75
B. €37.50
C. €35.00
D. €40.00
E. €42.50

7. Five patients — P, Q, R, S and T — are seen one after another. Q is seen immediately after P. R is seen before P. T is seen last. S is seen after Q. Who is seen second?

A. S
B. Q
C. P
D. T
E. R

8. What is the next number in the sequence: 3, 7, 15, 31, 63, ...?

A. 115
B. 127
C. 128
D. 95
E. 126

9. A tap fills a tank in 6 hours; a drain empties the full tank in 9 hours. If both are open, starting with an empty tank, how long does it take to fill the tank?

A. 3 hours
B. 15 hours
C. 12 hours
D. 7.5 hours
E. 18 hours

Section 3 — Biology (Q10–32)

10. Which organelle is the main site of synthesis of proteins destined for secretion?

A. Smooth endoplasmic reticulum
B. Peroxisome
C. Rough endoplasmic reticulum
D. Free cytosolic ribosomes
E. Lysosome

11. A segment of DNA template strand reads 3'-TACGGA-5'. What is the sequence of the mRNA transcribed from it?

A. 3'-AUGCCU-5'
B. 5'-UACGGA-3'
C. 5'-AUGCCU-3'
D. 5'-ATGCCT-3'
E. 5'-UCCGUA-3'

12. In which phase of mitosis do sister chromatids separate and move towards opposite poles?

A. Anaphase
B. Prometaphase
C. Metaphase
D. Prophase
E. Telophase

13. Which of the following processes produces the greatest number of ATP molecules per glucose in aerobic respiration?

A. Glycolysis
B. Krebs cycle (substrate-level phosphorylation)
C. Lactic acid fermentation
D. Link reaction
E. Oxidative phosphorylation

14. Red–green colour blindness is an X-linked recessive trait. A woman with normal vision whose father was colour-blind has children with a man with normal vision. What is the probability that their first son is colour-blind?

A. 3/4
B. 0
C. 1/4
D. 1/2
E. 1

15. Which molecule acts as the final electron acceptor in the mitochondrial electron transport chain?

A. Water
B. Cytochrome c
C. NAD^+
D. Oxygen
E. FAD

16. Which of the following is NOT a function of the liver?

A. Secretion of insulin
B. Synthesis of albumin
C. Production of bile
D. Storage of glycogen
E. Deamination of amino acids

17. Which statement about enzymes is correct?

A. Non-competitive inhibitors always bind to the active site.

B. A competitive inhibitor binds to the active site and its effect can be overcome by increasing substrate concentration.

C. Enzymes increase the activation energy of a reaction.

D. Enzymes change the equilibrium constant of a reaction.

E. Enzymes are consumed during the reaction.

18. During the depolarisation phase of a neuronal action potential, which event occurs?

A. The Na^+/K^+ pump moves 3 Na^+ in and 2 K^+ out.

B. The membrane potential becomes more negative.

C. K^+ rushes into the cell through voltage-gated channels.

D. Cl^- leaves the cell.

E. Na^+ rushes into the cell through voltage-gated channels.

19. Which blood vessel carries oxygenated blood from the lungs to the heart?

A. Coronary sinus

B. Pulmonary vein

C. Aorta

D. Pulmonary artery

E. Superior vena cava

20. In a dihybrid cross AaBb × AaBb with independent assortment and complete dominance, what fraction of offspring show both dominant phenotypes?

A. 3/16

B. 12/16

C. 9/16

D. 1/16

E. 6/16

21. Which hormone is released by the posterior pituitary?

A. Prolactin

B. Thyroid-stimulating hormone

C. Follicle-stimulating hormone

D. Antidiuretic hormone (ADH)

E. Growth hormone

22. Which type of bond holds the two strands of the DNA double helix together?

A. Hydrogen bonds between bases

B. Disulfide bonds

C. Glycosidic bonds

D. Phosphodiester bonds

E. Peptide bonds

23. Which of the following cells is responsible for producing antibodies?

A. Cytotoxic T cells

B. Plasma cells

C. Macrophages

D. Neutrophils

E. Helper T cells

24. In the nephron, where does most reabsorption of glucose normally occur?

A. Proximal convoluted tubule
B. Descending limb of the loop of Henle
C. Distal convoluted tubule
D. Collecting duct
E. Glomerulus

25. Which statement about meiosis is correct?

A. Crossing over occurs during prophase II.
B. It produces two genetically identical diploid cells.
C. It occurs in all somatic cells.
D. Homologous chromosomes separate during anaphase I.
E. DNA replicates between meiosis I and meiosis II.

26. Which of these is a polysaccharide used for energy storage in animals?

A. Sucrose
B. Chitin
C. Starch
D. Glycogen
E. Cellulose

27. A mutation changes a codon from UAC (Tyr) to UAA. What type of mutation is this?

A. Frameshift mutation
B. Insertion
C. Silent mutation
D. Missense mutation
E. Nonsense mutation

28. Which component of the plasma membrane is mainly responsible for regulating its fluidity in animal cells?

A. Integral proteins
B. Cellulose
C. Cholesterol
D. Glycolipids
E. Glycoproteins

29. Bacteria differ from eukaryotic cells because bacteria:

A. cannot contain plasmids.
B. always lack a cell wall.
C. have 70S ribosomes and lack membrane-bound organelles.
D. lack ribosomes.
E. have a nucleus surrounded by a double membrane.

30. Where in the human digestive system does most absorption of nutrients take place?

A. Large intestine
B. Stomach
C. Oesophagus
D. Duodenum only
E. Small intestine (jejunum and ileum)

31. Which of the following describes the role of tRNA in translation?

A. It brings a specific amino acid to the ribosome and pairs its anticodon with an mRNA codon.
B. It carries the genetic code from the nucleus to the cytoplasm.
C. It forms the structural core of the ribosome.
D. It removes introns from pre-mRNA.
E. It catalyses formation of phosphodiester bonds.

32. In a population in Hardy–Weinberg equilibrium, the frequency of a recessive phenotype is 0.16. What is the frequency of heterozygotes?

A. 0.24
B. 0.48
C. 0.60
D. 0.36
E. 0.16

Section 4 — Chemistry (Q33–47)

33. How many moles of oxygen atoms are present in 0.5 mol of $Ca(NO_3)_2$?

A. 2
B. 3
C. 1.5
D. 6
E. 9

34. What is the oxidation number of manganese in $KMnO_4$?

A. +5
B. +4
C. +2
D. +7
E. +6

35. What is the pH of a 0.001 mol/L solution of HCl at 25 °C?

A. 2
B. 1
C. 13
D. 11
E. 3

36. Which element has the highest first ionisation energy?

A. Ar
B. Cl
C. K
D. Mg
E. Na

37. What is the molecular geometry of NH_3?

A. Bent
B. Trigonal planar
C. Trigonal pyramidal
D. Linear
E. Tetrahedral

38. Balance the equation: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$. What is the coefficient of O_2 when the smallest whole-number coefficients are used?

A. 7
B. 5
C. 3
D. 10
E. 4

39. Which of the following compounds contains an ester functional group?

A. CH_3CH_2OH
B. CH_3COOCH_3
C. CH_3CHO
D. CH_3COCH_3
E. CH_3COOH

40. 20 g of NaOH (M = 40 g/mol) is dissolved in water to make 500 mL of solution. What is the concentration?

A. 4 mol/L
B. 1 mol/L
C. 0.5 mol/L
D. 2 mol/L
E. 0.25 mol/L

41. Which type of intermolecular force is mainly responsible for the high boiling point of water compared with H_2S?

A. Ionic bonding
B. Covalent bonding
C. London dispersion forces
D. Hydrogen bonding
E. Metallic bonding

42. Which of the following is a buffer solution?

A. $HNO_3 + KNO_3$
B. HCl + NaCl
C. $CH_3COOH + CH_3COONa$
D. NaOH + NaCl
E. $H_2SO_4 + Na_2SO_4$

43. For the exothermic equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, which change increases the yield of NH_3?

A. Adding a catalyst
B. Increasing pressure
C. Increasing temperature
D. Removing N_2
E. Decreasing pressure

44. How many structural isomers have the formula C_4H_{10}?

A. 1
B. 5
C. 3
D. 2
E. 4

45. An isotope has 17 protons and 20 neutrons. Which is it?

A. ^{37}Cl
B. ^{37}Ar
C. ^{34}S
D. ^{20}Ca
E. ^{37}K

46. In the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which species is the oxidising agent?

A. There is no oxidising agent.
B. Cu
C. Cu^{2+}
D. Zn
E. Zn^{2+}

47. Which of the following alkenes can show cis–trans (geometric) isomerism?

A. Ethene
B. Propene
C. 2-methylpropene
D. But-1-ene
E. But-2-ene

Section 5 — Physics & Mathematics (Q48–60)

48. A car accelerates uniformly from rest to 20 m/s in 5 s. How far does it travel in this time?

A. 200 m
B. 40 m
C. 20 m
D. 50 m
E. 100 m

49. A 2 kg object is lifted vertically by 3 m. Taking $g = 10\ m/s^2$, what is the gain in gravitational potential energy?

A. 6 J
B. 30 J
C. 600 J
D. 60 J
E. 15 J

50. Two resistors of 6 Ω and 3 Ω are connected in parallel. What is the equivalent resistance?

A. 9 Ω
B. 0.5 Ω
C. 18 Ω
D. 4.5 Ω
E. 2 Ω

51. A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

A. 200 m/s
B. 100 m/s
C. 12.5 m/s
D. 46 m/s
E. 54 m/s

52. What is the pressure exerted by a column of water 10 m deep ($\rho = 1000\ kg/m^3$, $g = 10\ m/s^2$), excluding atmospheric pressure?

A. 10^4 Pa
B. 10^3 Pa
C. 10^7 Pa
D. 10^6 Pa
E. 10^5 Pa

53. A body of mass 4 kg moving at 3 m/s has kinetic energy of:

A. 24 J
B. 36 J
C. 6 J
D. 12 J
E. 18 J

54. Which of the following is the SI unit of power?

A. $kg \cdot m^2/s^2$
B. $kg \cdot m/s^2$
C. $kg \cdot m^2/s^3$
D. $kg \cdot m/s$
E. $N \cdot s$

55. Solve for x: $\log_2(x) + \log_2(x - 2) = 3$.

A. 4
B. -2 and 4
C. 8
D. -2
E. 2

56. What is the value of $\sin(30°) \cdot \cos(60°) + \cos(30°) \cdot \sin(60°)$?

A. $\sqrt{3}/2$
B. 1
C. 1/2
D. 0
E. $\sqrt{3}$

57. Two fair six-sided dice are rolled. What is the probability that the sum is 7?

A. 1/9
B. 1/12
C. 7/36
D. 1/6
E. 1/36

58. What is the area of a circle whose circumference is 10π cm?

A. 5π cm²
B. 50π cm²
C. 25π cm²
D. 100π cm²
E. 10π cm²

59. The solutions of $x^2 - 5x + 6 = 0$ are:

A. $x = -2$ and $x = -3$
B. $x = 1$ and $x = 6$
C. $x = 2$ and $x = 3$
D. $x = -1$ and $x = 6$
E. $x = 5$ and $x = 6$

60. A current of 2 A flows through a resistor for 10 s. How much charge passes through it?

A. 5 C
B. 0.2 C
C. 12 C
D. 200 C
E. 20 C

정답 및 해설

번호	정답	해설
1	A	감염 증가가 없었다는 결과는 "방문객이 감염을 옮긴다"는 기존 근거를 지지하지 않음. B는 "감소"까지 주장해 과도.
2	A	비판은 "연구 규모가 작고 업무량 미측정". 대규모·업무량 측정 연구가 차이 없음을 보이면 비판이 약화.
3	E	윌리엄 하비(1628, *De Motu Cordis*).
4	D	로마조약 → 유럽경제공동체(EEC) 설립.
5	A	이탈리아어를 하는 의사는 야간근무 안 함 → 야간근무 안 하는 의사가 존재.
6	B	40×0.75=30, 30×1.25=37.5.
7	C	T 마지막, R<P, P 바로 뒤 Q, S는 Q 뒤 → R, P, Q, S, T. 두 번째는 P.
8	B	각 항 = 2×이전+1 → 127.
9	E	1/6 − 1/9 = 1/18 → 18시간.
10	C	분비 단백질은 조면소포체 리보솜에서 합성.
11	C	주형 3'-TACGGA-5' → mRNA 5'-AUGCCU-3'.
12	A	후기(anaphase)에 자매염색분체 분리.
13	E	산화적 인산화가 ATP 대부분(~26–28개) 생성.
14	D	어머니는 보인자(X^AX^a) → 아들 1/2.
15	D	최종 전자수용체는 O_2.
16	A	인슐린은 췌장 β세포.
17	B	경쟁적 저해는 기질 농도↑로 극복 가능.
18	E	탈분극: 전압개폐 Na^+ 통로로 Na^+ 유입.
19	B	폐정맥이 산소 풍부한 혈액을 좌심방으로.
20	C	9:3:3:1 → 9/16.
21	D	후엽: ADH, 옥시토신.
22	A	두 가닥은 염기 간 수소결합.
23	B	형질세포(분화된 B세포)가 항체 분비.
24	A	포도당은 근위세뇨관에서 거의 전부 재흡수.
25	D	감수 1분열 후기에 상동염색체 분리.
26	D	동물의 저장 다당류 = 글리코겐.
27	E	UAA는 종결코돈 → 넌센스 돌연변이.
28	C	콜레스테롤이 막 유동성 조절.
29	C	원핵세포: 70S 리보솜, 막성 세포소기관 없음.
30	E	소장(공장·회장)에서 대부분 흡수.
31	A	tRNA는 아미노산 운반, 안티코돈-코돈 결합.
32	B	$q^2=0.16$ → q=0.4, p=0.6, 2pq=0.48.
33	B	1 mol당 O 6개 → 0.5×6=3.
34	D	K(+1)+Mn+4(−2)=0 → +7.

번호	정답	해설
35	E	$[H^+]=10^{-3}$ → pH 3.
36	A	비활성기체 Ar이 가장 높음.
37	C	비공유전자쌍 1개 → 삼각뿔.
38	B	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.
39	B	아세트산메틸(에스터).
40	B	0.5 mol / 0.5 L = 1 mol/L.
41	D	물의 수소결합.
42	C	약산 + 그 짝염기.
43	B	기체 몰수 4→2, 압력↑ 시 정반응. 촉매는 수율 불변.
44	D	부탄, 2-메틸프로페인.
45	A	양성자 17 = Cl, 질량수 37.
46	C	Cu^{2+}가 환원됨 → 산화제.
47	E	부트-2-엔만 각 이중결합 탄소에 서로 다른 치환기.
48	D	평균속도 10 m/s × 5 s = 50 m.
49	D	$mgh = 2\times10\times3 = 60$ J.
50	E	$6\cdot3/(6+3) = 2\ \Omega$.
51	A	$v = f\lambda = 200$ m/s.
52	E	$\rho gh = 1000\times10\times10 = 10^5$ Pa.
53	E	$\frac{1}{2}\cdot4\cdot9 = 18$ J.
54	C	$W = J/s = kg\cdot m^2/s^3$.
55	A	$x(x-2)=8 \rightarrow x=4$ ($x=-2$는 정의역 밖).
56	B	$\sin(30°+60°) = \sin90° = 1$.
57	D	6/36 = 1/6.
58	C	r = 5 → 25π.
59	C	$(x-2)(x-3)=0$.
60	E	$Q = It = 20$ C.